

CS & IT ENGINEERING



Operating System

Basics of OS

One Shot



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Topics to be Covered



Topic

Operating System Definition

Topic

Types of Operating System

Topic

Dual Mode of Operation



Topic : Introduction

❑ **GATE Ranks:**

- 682 (2009) – 3rd year
- 19 (2010) – 4th year
- 119,440 etc.

❑ **Education:**

- ME from IISc Bangalore
- M. tech from BITS-pilani in Data Science

❑ **Work:**

- 18+ Year Teaching Experience
- 15+ in GATE/IES
- Worked in Cisco, Audience Communication



Topic : Operating System



Interface between user and hardware





Topic : Operating System

- Set of utilities to simplify application development/execution

↳ API



Topic : OS Course Syllabus

Chapter Number	Chapter Name
1	Introduction
2	Process Management
3	CPU Scheduling
4	Process Synchronization
5	Deadlock
6	Memory Management & Virtual Memory
7	File System
8	Disk Scheduling



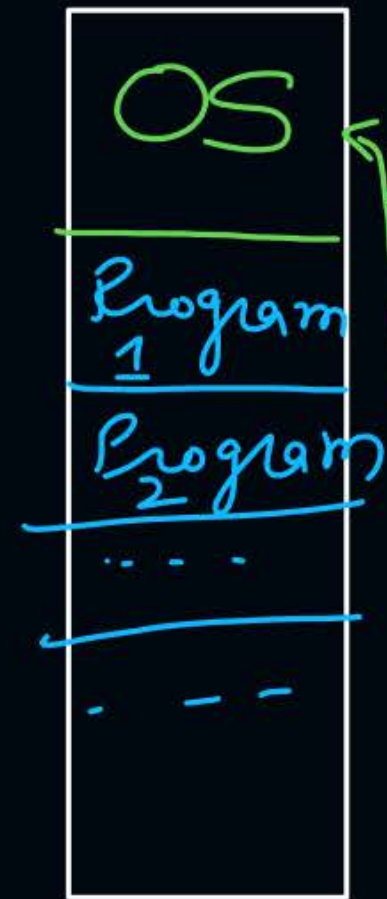
Topic : Services of OS



- User Interface
- Program Execution *→ most important*
- I/O Operation ✓
- File-System Manipulation ✓
- Communication (Inter-process Communication) ✓
- Error Detection ✓
- Resource Allocation ✓
- Accounting ✓
- Protection & Security /

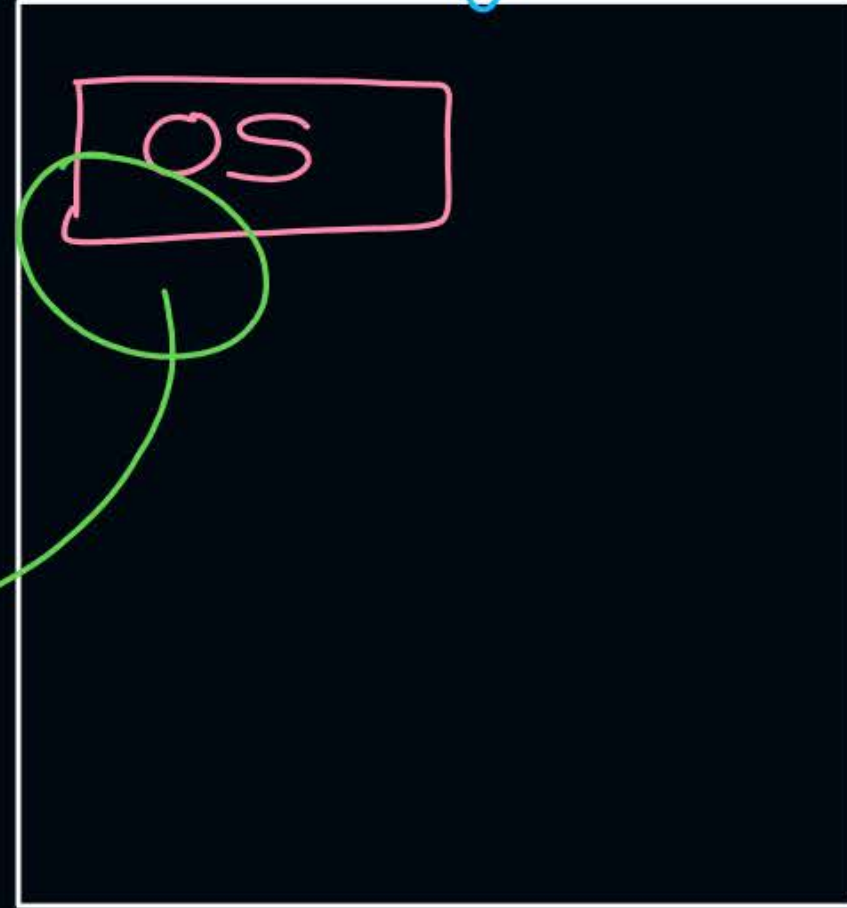


(memory or main mem.
RAM



or
Primary
or
physical
memory)

secondary mem.





Topic : Types of OS



1. Uniprogramming OS
2. Multiprogramming OS
3. Multitasking OS (Time Sharing)
4. Multiprocessing OS
5. Multiuser OS
6. Real Time OS
7. Embedded OS
8. Handheld Device OS



Topic : Uniprogramming OS

This OS allows only one program to be in main memory at a time.

Not a great utilization of CPU.

CPU

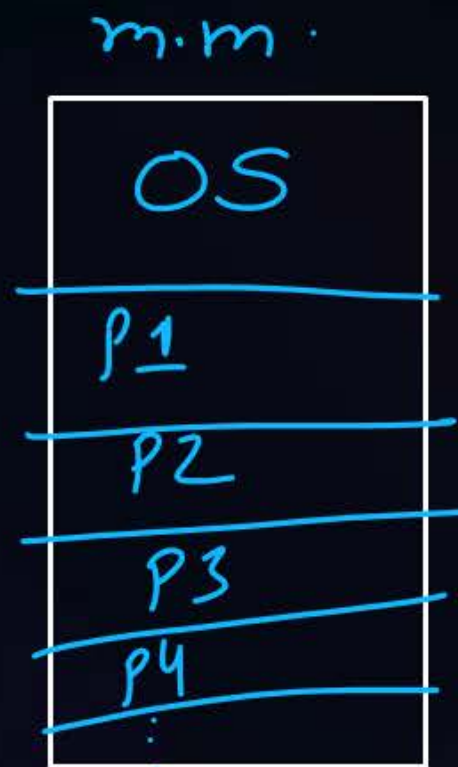


single program was not able to utilize CPU because if the program goes for I/O operation then CPU will be free.



Topic : Multiprogramming OS

This OS allows more than one programs to be in main memory at a time.



If one program leaves CPU and goes for I/O operation then OS can schedule other program to run on CPU to utilize CPU.

Degree of multiprogramming:-

No. of programs in mm at a time.

as degree of multiprogramming increases, the CPU utilization also increases, but upto a certain level.

Types of multiprogramming

```
graph TD; A[Types of multiprogramming] --> B[Non-preemptive]; A --> C[Preemptive];
```

Non-preemptive

A current running process can leave CPU only on its wish, and can not be taken out of CPU forcefully.

Preemptive

A running program can be taken out of CPU forcefully (against program's wish)



Topic : Multitasking OS (Time-shared OS)

It is an extension of multiprogramming OS, in which programs are executed on CPU in round-robin manner.

CPU



very-very
small to be
observed by
human



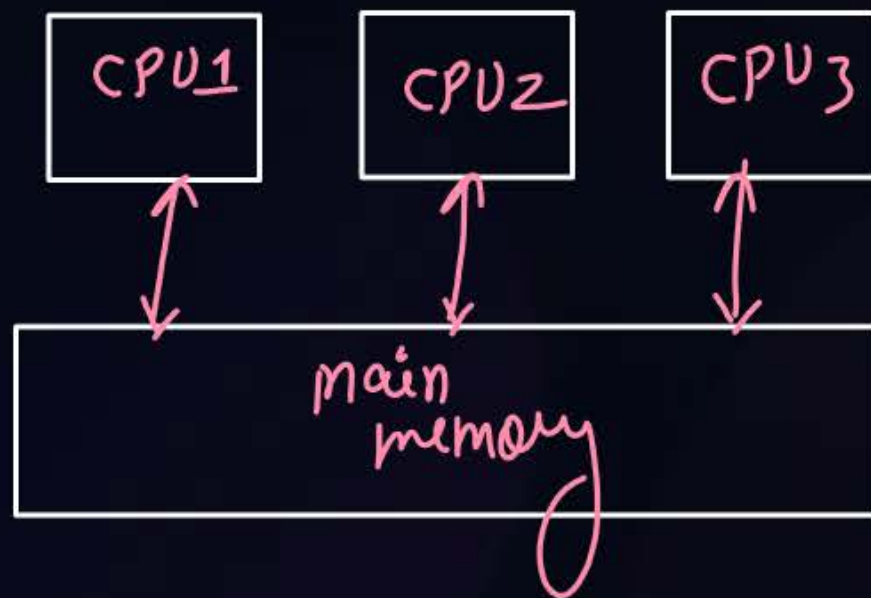


Topic : Multiprocessing OS

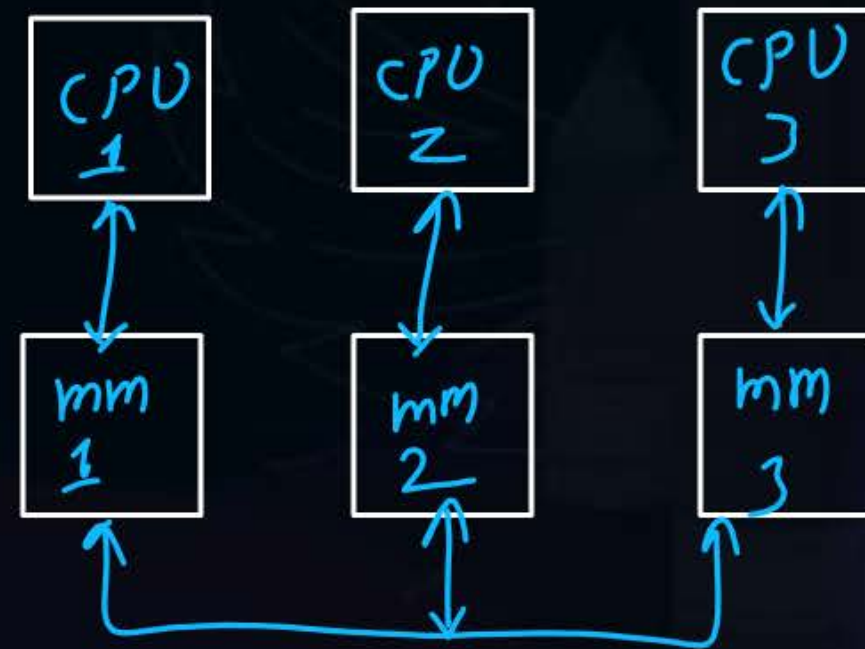
This OS works on computers with multiple CPUs.

Types

Tightly coupled
or
shared memory



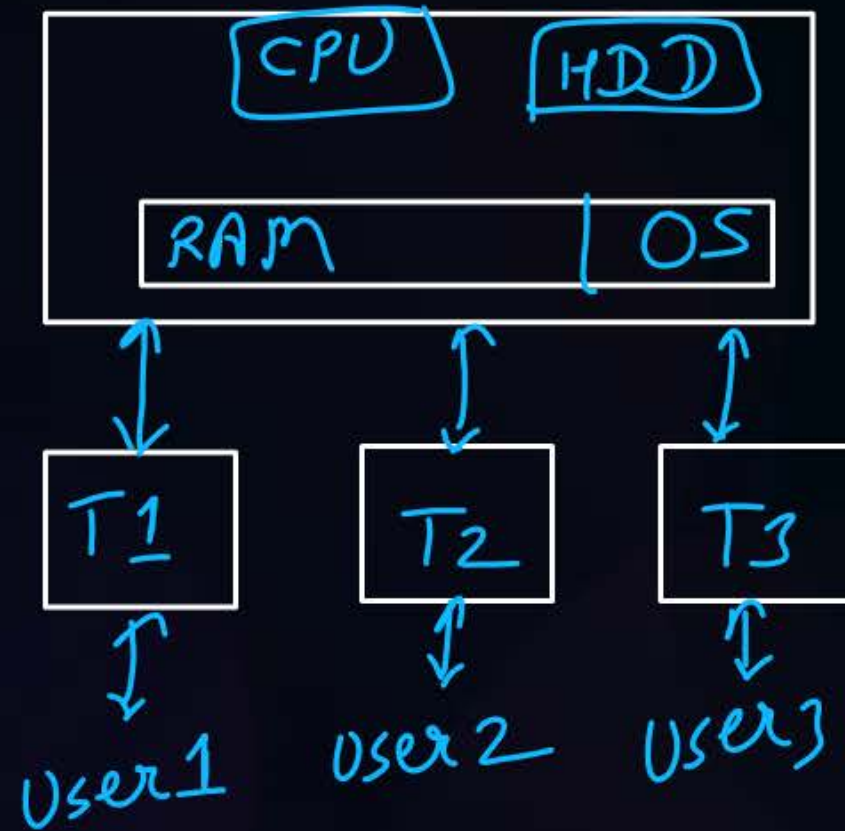
Loosely coupled
or
distributed





Topic : Multiuser OS

This OS allows multiple users to use single computer system simultaneously.



$T_1, T_2, T_3 \Rightarrow$ Terminals
(screen, keyboard, mouse
& I/O devices)

Apart from MS OS and apple OS
all are multiuser.



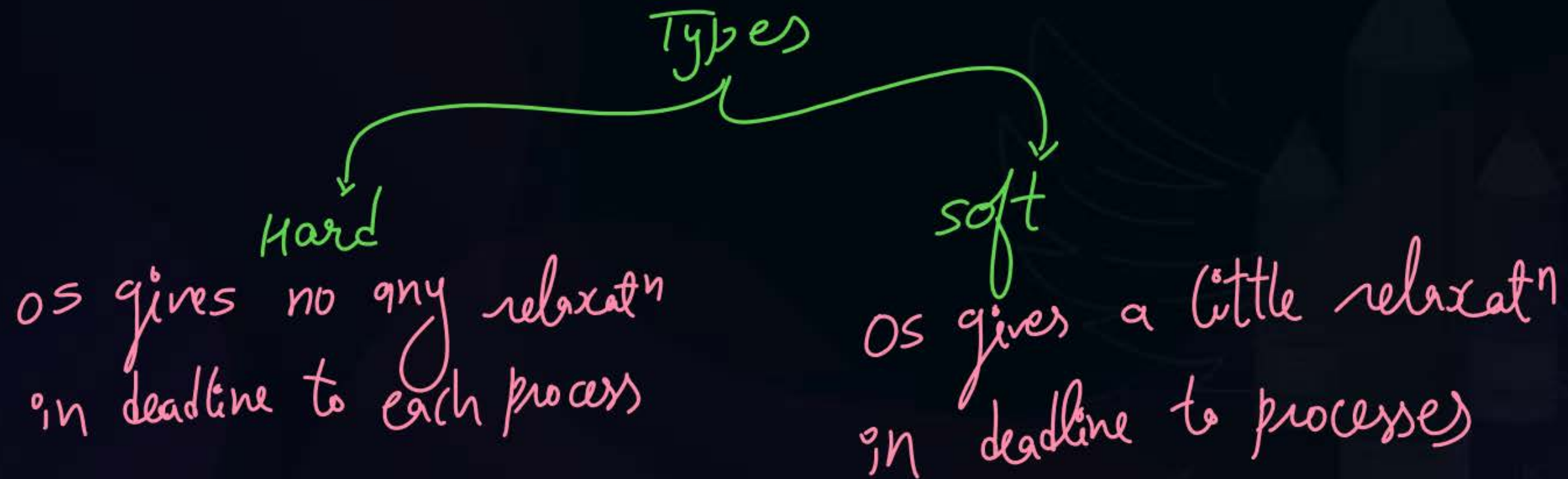
Topic : Real Time OS



OS works on real time data and event.

example:- Rocket launch

⇒ each program has a deadline.





Topic : Embedded OS



The s/w part given with the embedded system.

↓
The Computing embedded with
devices like car, AC, washing m/c.



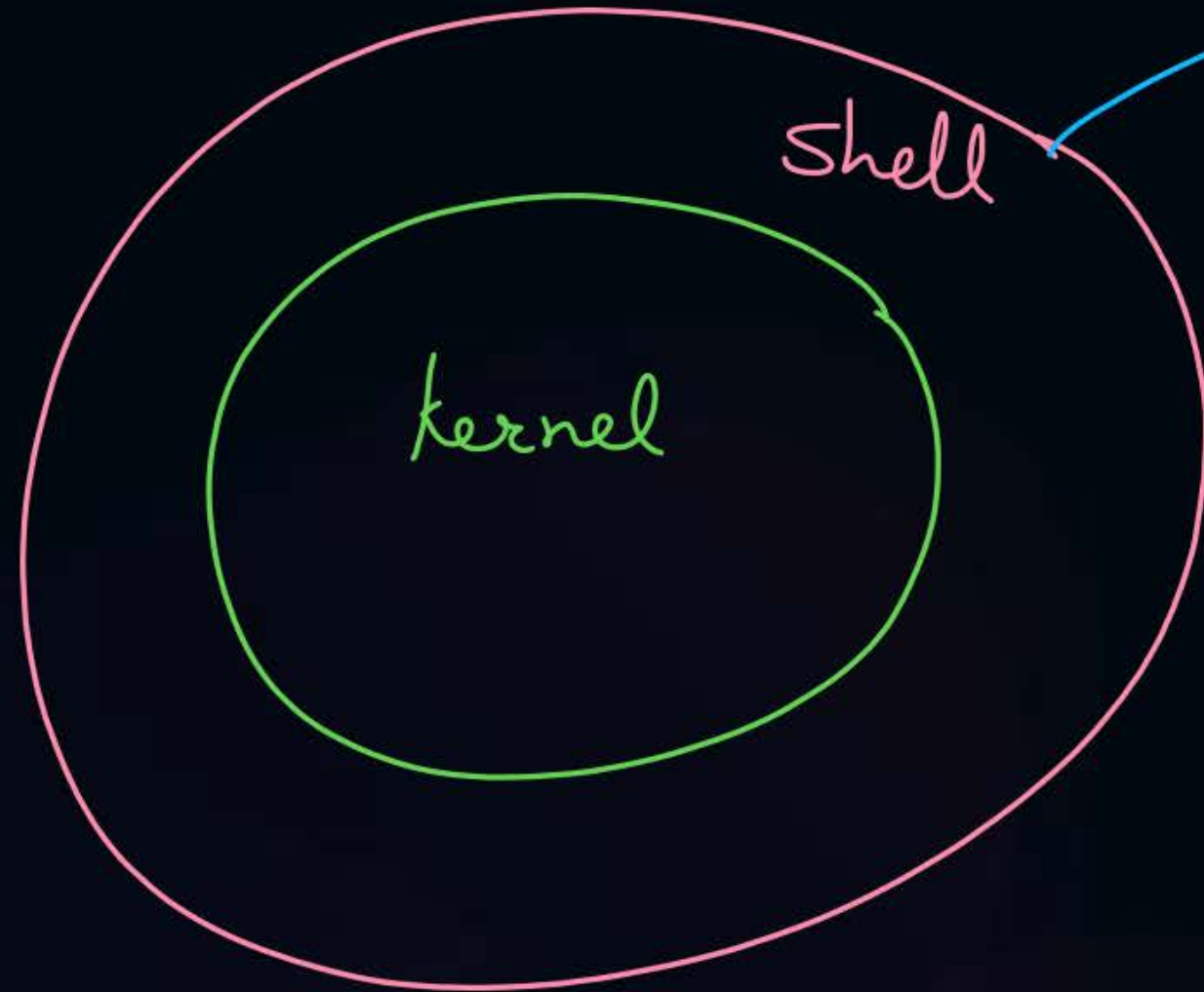
Topic : Handheld Device OS

↓
works on hand held devices .





Topic : Parts of OS



→ GUI (Graphical User Interface)
→ Command Line Interpreter





2 mins Summary

Topic

Operating System Definition

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Types of Operating System

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Dual Mode of Operation



Happy Learning

THANK - YOU

